Cellcraft: Cell Differentiation

Grade 7 Life Science | STUDENT WORKSHEET

Name: _____ Date: _____

Open the app's Worksheet tab. Follow questions 1-10. Play each linked activity, then record your observations here. Use short, clear answers.

1. A cell develops a specialty

APP: Field guide

In your own words, what happens to a cell during differentiation?

2. Nerve cell: send a signal

APP: Nerve cell

How do a nerve cell's long extension and branches help it do its job?

3. Muscle cell: lift the load

APP: Muscle cell

In the lifting model, what change helps the muscle cell pull? Why does contraction need a supply of energy?

4. Skin cells: build a barrier

APP: Skin surface cell

How do closely packed skin epithelial cells help protect the body?

5. Red blood cell: carry oxygen

APP: Red blood cell

What does hemoglobin carry? Does a mature human red blood cell have a nucleus?

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Continue in the app: Leaf cell, Root hair cell, Field guide, and Tissue tower.

6. Leaf cell: use light energy

APP: Leaf cell

Complete this word model of photosynthesis. Name the two materials used and the gas released.

_____ + _____ + light energy -> sugar + _____

This is a word model, not a balanced chemical equation.

7. Root hair: absorb water

APP: Root hair cell

How does a root hair's long extension help absorption? Name the process by which water enters.

8. Same DNA, different jobs

APP: Field guide

Why can a nerve cell and a muscle cell in the same person do different jobs even though they usually contain the same DNA?

9. More cells or a new specialty?

APP: Field guide

Write division or differentiation for each event.

a. One cell becomes two similar cells. _____

b. A developing cell forms nerve-cell structures. _____

10. Tissue tower: move your arm

APP: Tissue tower

Use the word bank to complete the five levels from smallest to largest.

Word bank: Person | Biceps muscle | Skeletal muscle tissue | Muscular system | Skeletal muscle cell

1. Cell: _____
2. Tissue: _____
3. Organ: _____
4. Organ system: _____
5. Organism: _____

Use short sentences. Explain how a structure helps a cell perform its job.